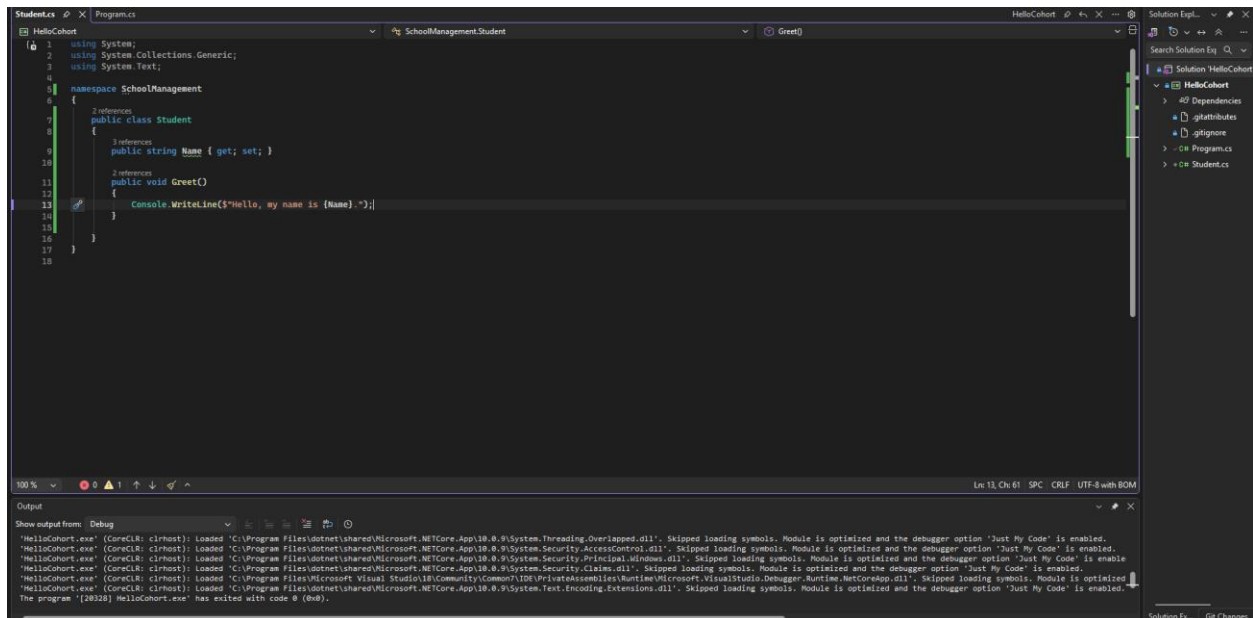


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Procedure:

I have created a new file called “Student.cs”. Inside Student.cs I have created a namespace called SchoolManagement with Student Class. This class contains the variable Name and method Greet to display the name of Student. Inside Program.cs we create two objects student1 and student2 and assign name and call greet function with and without using directives.

Student.cs file:



The screenshot shows the Visual Studio IDE with the 'Student.cs' file open. The code defines a namespace 'SchoolManagement' containing a 'Student' class. The 'Student' class has a 'Name' property and a 'Greet()' method. The 'Greet()' method uses 'Console.WriteLine' to display the name. The 'Program.cs' file is also visible, showing the 'Greet()' method being called. The output window at the bottom shows the execution of the program, displaying the message 'Hello, my name is {Name}.'.

```

1 using System;
2 using System.Collections.Generic;
3 using System.Text;
4
5 namespace SchoolManagement
6 {
7     2 references
8     public class Student
9     {
10         3 references
11         public string Name { get; set; }
12
13         2 references
14         public void Greet()
15         {
16             Console.WriteLine($"Hello, my name is {Name}.");
17         }
18     }
19 }

```

Output

```

Show output from: Debug
'HelloCohort.exe' (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\10.0.0\System.Threading.Overlapped.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
'HelloCohort.exe' (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\10.0.0\System.Security.AccessControl.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
'HelloCohort.exe' (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\10.0.0\System.Security.Principal.Windows.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
'HelloCohort.exe' (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\10.0.0\System.Security.Claims.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
'HelloCohort.exe' (CoreCLR: clrhost): Loaded 'C:\Program Files\Microsoft Visual Studio\10\Community\Common7\IDE\PrivateAssemblies\Microsoft.VisualStudio.Debugger.Runtime\NetCoreApp.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
The program '[28028] HelloCohort.exe' has exited with code 0 (0x0).

```

Program.cs file:

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```

1 using SchoolManagement;
2
3 class Program
4 {
5     static void Main(string[] args)
6     {
7         //Without using directive -> must write full namespace path
8         var student1 = new SchoolManagement.Student();
9         student1.Name = "Ravi";
10        student1.Greet();
11
12        //With using directive -> can directly use class name
13        var student2 = new Student();
14        student2.Name = "Rithika";
15        student2.Greet();
16    }
17 }
18

```

Output

```

Show output from: Debug
'HelloCohort.exe' (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\10.0.9\System.Threading.Overlapped.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
'HelloCohort.exe' (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\10.0.9\System.Security.AccessControl.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
'HelloCohort.exe' (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\10.0.9\System.Security.Principal.Windows.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
'HelloCohort.exe' (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\10.0.9\System.Security.Claims.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
'HelloCohort.exe' (CoreCLR: clrhost): Loaded 'C:\Program Files\Microsoft Visual Studio\10\Community\Common7\IDE\PrivateAssemblies\Runtime\Microsoft.VisualStudio.Debugger.Runtime.NetCoreApp.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
'HelloCohort.exe' (CoreCLR: clrhost): Loaded 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App\10.0.9\System.Text.Encoding.Extensions.dll'. Skipped loading symbols. Module is optimized and the debugger option 'Just My Code' is enabled.
The program '[20328] HelloCohort.exe' has exited with code 0 (0x0).

```

Output:

```

Microsoft Visual Studio Debug Console
Hello, my name is Ravi.
Hello, my name is Rithika.

C:\Users\ravi.rithika\HelloCohort\HelloCohort\bin\Debug\net10.0\HelloCohort.exe (process 20328) exited with code 0 (0x0)
.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .|

```

Task 1.6 Create `ModuleA` and `ModuleB`, each with `Helper.Greet()`. Call both from `Main()` and resolve the name clash with fully-qualified names. ✓ Done when: both `Greet()`s run without ambiguity errors.

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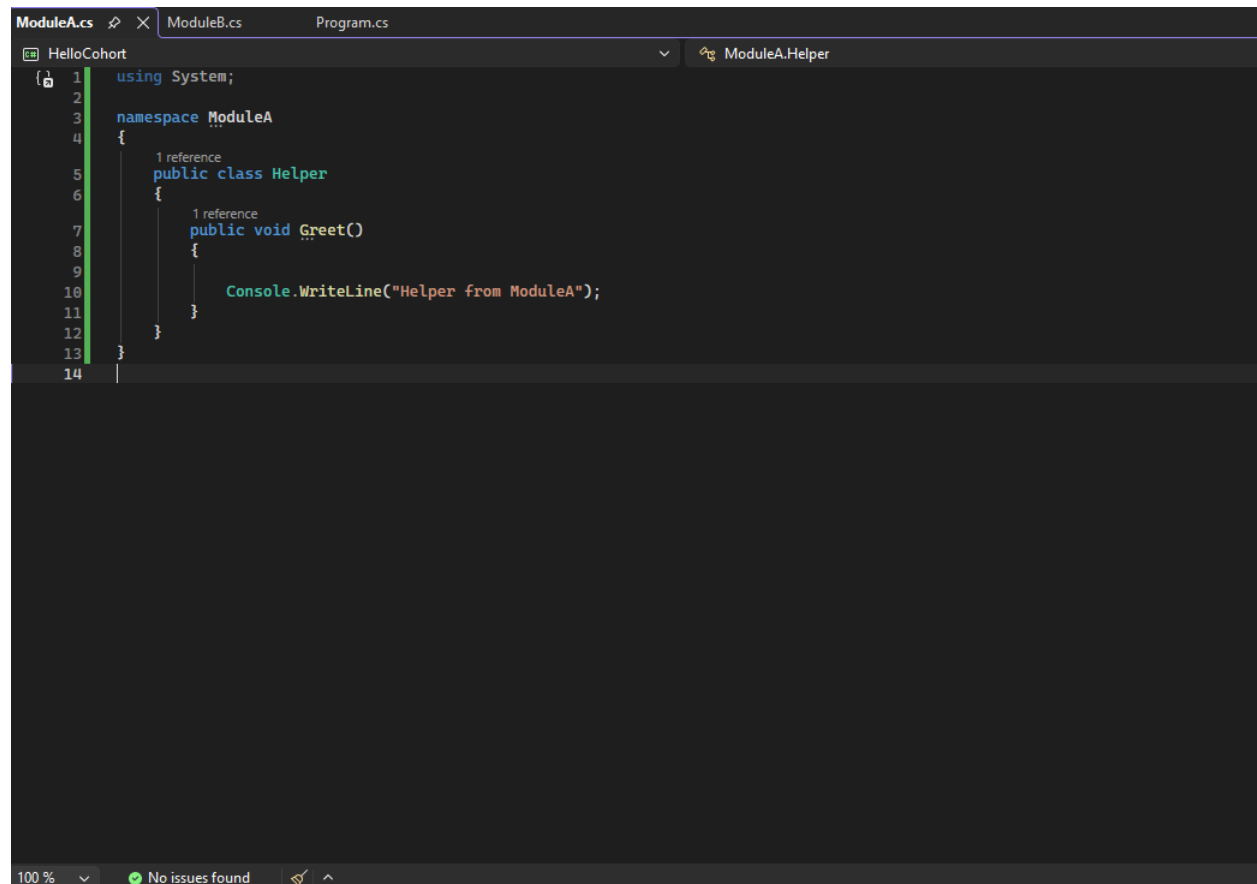
Procedure:

I have created two files Module A and Module B where both modules have the same class Helper and a greet function with different print statements. There are two options to this name conflict

- Use fully qualified names (ModuleName.Helper.MethodName)
- Use aliases (using namespace = ModuleName.Helper)

Output:

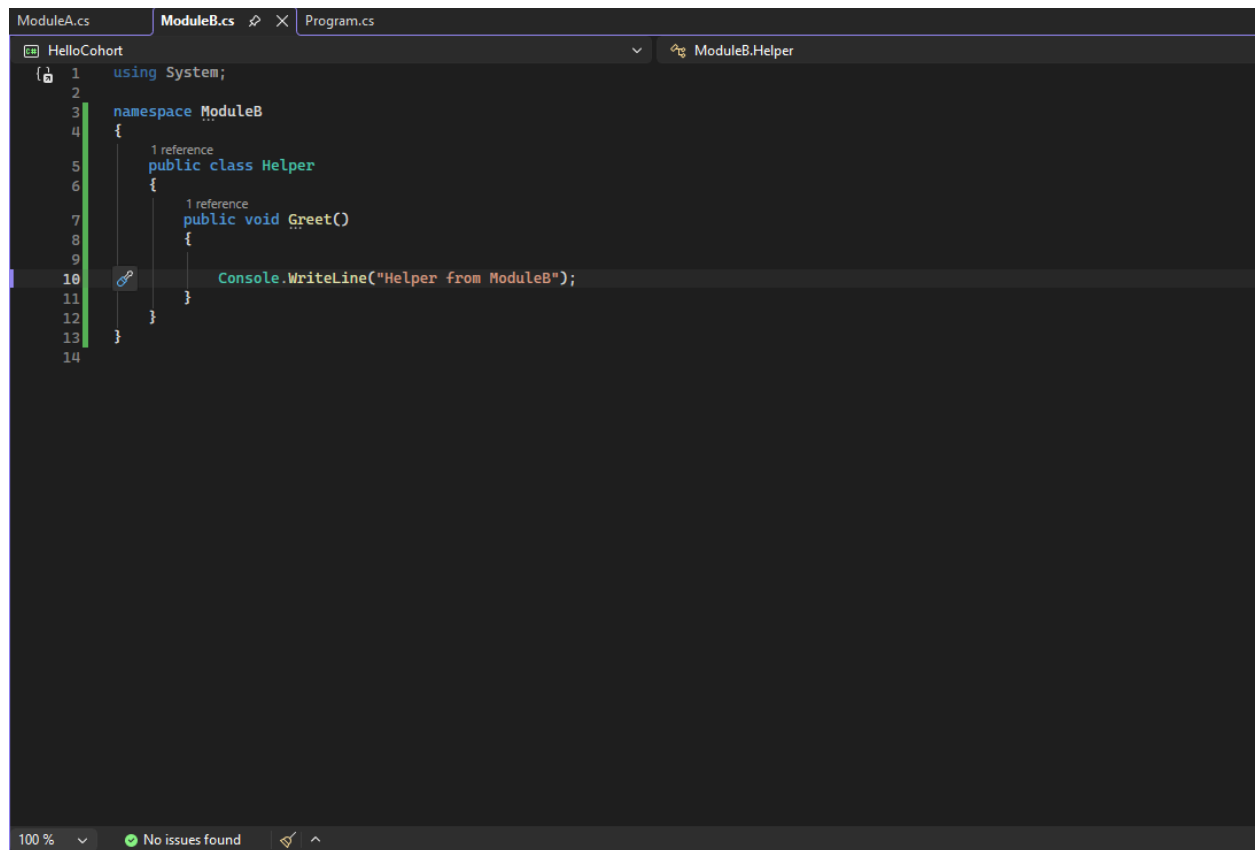
ModuleA file:



```
ModuleA.cs | ModuleB.cs | Program.cs
HelloCohort
1 using System;
2
3 namespace ModuleA
4 {
5     1 reference
6     public class Helper
7     {
8         1 reference
9         public void Greet()
10        {
11            Console.WriteLine("Helper from ModuleA");
12        }
13    }
14 }
```

ModuleB file:

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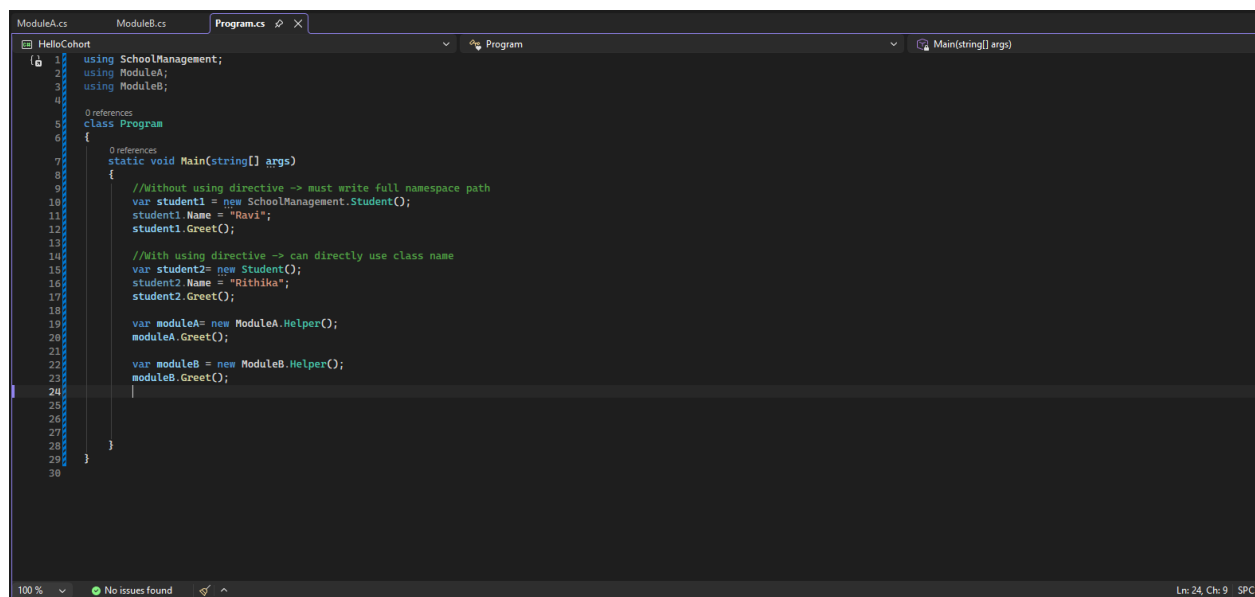


The screenshot shows the Visual Studio IDE with three tabs: ModuleA.cs, ModuleB.cs (active), and Program.cs. The active file, ModuleB.cs, contains the following code:

```
1 using System;
2
3 namespace ModuleB
4 {
5     1 reference
6     public class Helper
7     {
8         1 reference
9         public void Greet()
10        {
11            Console.WriteLine("Helper from ModuleB");
12        }
13    }
14 }
```

The status bar at the bottom indicates "100 %", "No issues found", and a zoom icon.

Program.cs:



The screenshot shows the Visual Studio IDE with three tabs: ModuleA.cs, ModuleB.cs, and Program.cs (active). The active file, Program.cs, contains the following code:

```
1 using SchoolManagement;
2 using ModuleA;
3 using ModuleB;
4
5 0 references
6 class Program
7 {
8     0 references
9     static void Main(string[] args)
10    {
11        //Without using directive -> must write full namespace path
12        var student1 = new SchoolManagement.Student();
13        student1.Name = "Ravi";
14        student1.Greet();
15
16        //With using directive -> can directly use class name
17        var student2 = new Student();
18        student2.Name = "Rithika";
19        student2.Greet();
20
21        var moduleA = new ModuleA.Helper();
22        moduleA.Greet();
23
24        var moduleB = new ModuleB.Helper();
25        moduleB.Greet();
26    }
27 }
28
29
30
```

The status bar at the bottom indicates "100 %", "No issues found", and a zoom icon. The right side of the status bar shows "Ln: 24, Ch: 9 SPC".

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```

Microsoft Visual Studio Debug Console
Hello, my name is Ravi.
Hello, my name is Rithika.
Helper from ModuleA
Helper from ModuleB

C:\Users\ravi.rithika\HelloCohort\HelloCohort\bin\Debug\net10.0\HelloCohort.exe (process 620) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```

Alternative Method:

```

ModuleA.cs  ModuleB.cs  Program.cs
HelloCohort
1 using SchoolManagement;
2 using ModuleA;
3 using ModuleB;
4 using HelperA = ModuleA.Helper;
5 using HelperB = ModuleB.Helper;
6
7 // references
8 class Program
9 {
10     // references
11     static void Main(string[] args)
12     {
13         //Without using directive -> must write full namespace path
14         var student1 = new SchoolManagement.Student();
15         student1.Name = "Ravi";
16         student1.Greet();
17
18         //With using directive -> can directly use class name
19         var student2 = new Student();
20         student2.Name = "Rithika";
21         student2.Greet();
22
23         var moduleA = new ModuleA.Helper();
24         moduleA.Greet();
25
26         var moduleB = new ModuleB.Helper();
27         moduleB.Greet();
28
29         // Alternatively, use aliases:
30
31         var HelperA = new HelperA();
32         HelperA.Greet();
33
34         var HelperB = new HelperB();
35         HelperB.Greet();
36
37
38
39
40
41
42
43
44
45
46
47
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49
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54
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99
100

```

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```

Microsoft Visual Studio Debug Console
Hello, my name is Ravi.
Hello, my name is Rithika.
Helper from ModuleA
Helper from ModuleB
Helper from ModuleA
Helper from ModuleB

C:\Users\ravi.rithika\HelloCohort\HelloCohort\bin\Debug\net10.0\HelloCohort.exe (process 19220) exited with code 0 (0x0)
.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```

Task 1.7 Declare 5 variables using correct conventions (camelCase locals, PascalCase types/methods). Try a variable named `class`, observe the error, fix with `@class`. ✓ Done when: the `@class` version compiles and the error is noted.

Procedure:

Declared 5 variables using camelCase `firstName`, `studentAge`, `averageScore`, `isEnrolled`, `gradeLevel`. Then used PascalCase for `PrintSummaryMethod`. Tried initializing `int class=1` and got CS1001 and CS1002 Identifier expected errors. After observing fixed the error with “`@class`”

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```
ModuleA.cs  ModuleB.cs  Program.cs  x
HelloCohort  Program  Main(string[] args)
1  using SchoolManagement;
2  using ModuleA;
3  using ModuleB;
4  using HelpA = ModuleA.Helper;
5  using HelpB = ModuleB.Helper;
6
7  0 references
8  class Program
9  {
10     0 references
11     static void Main(string[] args)
12     {
13         //Declare 5 variables using correct conventions
14         string firstName = "Rithika";
15         int studentAge = 24;
16         double averageScore = 99.5;
17         bool isEnrolled = true;
18         char gradeLevel = 'A';
19
20         //PascalCase for types and methods
21         void PrintSummary()
22         {
23             Console.WriteLine($"{firstName}, age {studentAge}, grade {gradeLevel}");
24         }
25         PrintSummary();
26
27
28
29
30
31         /*
32         //Without using directive -> must write full namespace path
33         var student1 = new SchoolManagement.Student();
34         student1.Name = "Ravi";
35         student1.Greet();
36
37         //With using directive -> can directly use class name
38         var student1 = new Student();
39     }
40 }
```

```
Microsoft Visual Studio Debug Console
Rithika, age 24, grade A

C:\Users\ravi.rithika\HelloCohort\HelloCohort\bin\Debug\net10.0\HelloCohort.exe (process 25424) exited with code 0 (0x0)
.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .|
```

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```

ModuleA.cs  ModuleB.cs  Program.cs  x
HelloCohort  Program
4  using HelpA = ModuleA.Helper;
5  using HelpB = ModuleB.Helper;
6
7  0 references
  class Program
8  {
9      0 references
      static void Main(string[] args)
10     {
11
12         //Declare 5 variables using correct conventions
13         string firstName = "Rithika";
14         int studentAge = 24;
15         double averageScore= 99.5;
16         bool isEnrolled = true;
17         char gradeLevel = 'A';
18
19         //PascalCase for types and methods
20         void PrintSummary()
21         {
22             Console.WriteLine($"{firstName}, age {studentAge}, grade {gradeLevel}");
23         }
24         PrintSummary();
25
26         // — Keyword clash demo —
27         int class = 1;
28         // CS1001 Identifier expected
29         // CS1002 ; expected
30         // 'class' is a reserved keyword; the parser gets confused.
31
32         //int @class = 1; // ✓ @ tells the compiler "treat 'class' as an identifier"
33         //Console.WriteLine($"class variable value: {@class}");
34
35
36
37
38         /*
39         //Without using directive -> must write full namespace path
40         var student1 = new SchoolManagement.Student();
41         student1.Name = "David";
42     }
43 }
100% 8 2

```


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ModuleA.csModuleB.csProgram.cs

Program.cs

```
14 int studentAge = 24;
15 double averageScore = 99.5;
16 bool isEnrolled = true;
17 char gradeLevel = 'A';
18
19 //PascalCase for types and methods
20 void PrintSummary()
21 {
22     Console.WriteLine($"{firstName}, age {studentAge}, grade {gradeLevel}");
23 }
24 PrintSummary();
25
26 // — Keyword clash demo —
27 int class = 1;
28 // CS1001 Identifier expected
29 // CS1002 ; expected
30 // 'class' is a reserved keyword; the parser gets confused.
31
32 //int @class = 1; // \ @ tells the compiler "treat 'class' as an identifier"
33 //Console.WriteLine($"class variable value: {@class}");
34
35
36
37
38 /*
39 //Without using directive -> must write full namespace path
40 var student1 = new SchoolManagement.Student();
41 student1.Name = "Ravi";
42 student1.Greet();
43
44 //With using directive -> can directly use class name
45 var student2 = new Student();
46 student2.Name = "Rithika";
47 student2.Greet();
48
49 var moduleA = new ModuleA.Helper();
50 moduleA.Greet();
51
52 var moduleB = new ModuleB.Helper();
53 moduleB.Greet();
```

100% 8 2 0 of 4 Messages Build • IntelliSense Lw: 27, Ch: 13 SPC CRL

Error List

Entire Solution 8 Errors 2 Warnings 0 of 4 Messages Build • IntelliSense Search Error List

Code	Description	Project	File
CS1001	Identifier expected	HelloCohort	Program.cs
CS1002	; expected	HelloCohort	Program.cs
CS1513	) expected	HelloCohort	Program.cs
CS1001	Identifier expected	HelloCohort	Program.cs
CS1514	{ expected	HelloCohort	Program.cs
CS1513	) expected	HelloCohort	Program.cs
CS1519	Invalid token '=' in a member declaration	HelloCohort	Program.cs
CS1002	Type or namespace definition, or end-of-file expected	HelloCohort	Program.cs
CS0219	The variable 'averageScore' is assigned but its value is never used	HelloCohort	Program.cs
CS0219	The variable 'isEnrolled' is assigned but its value is never used	HelloCohort	Program.cs

Error List Output

Program.csMain(string[] args)

```
11 //Declare 5 variables using correct conventions
12 string firstName = "Rithika";
13 int studentAge = 24;
14 double averageScore = 99.5;
15 bool isEnrolled = true;
16 char gradeLevel = 'A';
17
18 //PascalCase for types and methods
19 void PrintSummary()
20 {
21     Console.WriteLine($"{firstName}, age {studentAge}, grade {gradeLevel}");
22 }
23 PrintSummary();
24
25 // — Keyword clash demo —
26 //int class = 1;
27 // CS1001 Identifier expected
28 // CS1002 ; expected
29 // 'class' is a reserved keyword; the parser gets confused.
30
31 int @class = 1; // \ @ tells the compiler "treat 'class' as an identifier"
32 Console.WriteLine($"class variable value: {@class}");
33
34
35
36
37
38 /*
39 //Without using directive -> must write full namespace path
40 var student1 = new SchoolManagement.Student();
41 student1.Name = "Ravi";
42 student1.Greet();
43
44 //With using directive -> can directly use class name
45 var student2 = new Student();
46 student2.Name = "Rithika";
47 student2.Greet();
48
49 var moduleA = new ModuleA.Helper();
50 moduleA.Greet();
51
52 var moduleB = new ModuleB.Helper();
53 moduleB.Greet();
```

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```

Rithika, age 24, grade A
class variable value: 1

C:\Users\ravi.rithika\HelloCohort\HelloCohort\bin\Debug\net10.0\HelloCohort.exe (process 12428) exited with code 0 (0x0)
.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```

Task 1.8 Use `#define TRIAL_VERSION` + `#if/#else/#endif` to switch a printed message; use `#region` to group a class into Fields/Properties/Constructors/Methods. ✓ Done when: toggling the `#define` changes the output.

Procedure:

I used `#define TRIAL_VERSION` + `#if/#else/#endif` to switch a printed message in `program.cs` and then use `#region` to group a class into Fields/Properties/Constructors/Methods for better code structure and readability.

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```

Course.cs  ModuleA.cs  ModuleB.cs  Program.cs
HelloCohort
1  #define TRIAL_VERSION
2  using SchoolManagement;
3  using ModuleA ;
4  using ModuleB;
5  using HelpA = ModuleA.Helper;
6  using HelpB = ModuleB.Helper;
7
8  0 references
9  class Program
10 {
11     0 references
12     static void Main(string[] args)
13     {
14         //Declare 5 variables using correct conventions
15         string firstName = "Rithika";
16         int studentAge = 24;
17         double averageScore = 99.5;
18         bool isEnrolled = true;
19         char gradeLevel = 'A';
20
21         //PascalCase for types and methods
22         void PrintSummary()
23         {
24             Console.WriteLine($"{firstName}, age {studentAge}, grade {gradeLevel}");
25         }
26         PrintSummary();
27
28         // — Keyword clash demo —
29         //int class = 1;
30         // CS1001 Identifier expected
31         // CS1002 ; expected
32         // 'class' is a reserved keyword; the parser gets confused.
33
34         int @class = 1; // / @ tells the compiler "treat 'class' as an identifier"
35         Console.WriteLine($"class variable value: {@class}");
36
37         //Preprocessor directives
38         Console.WriteLine("App starting...");
39
40         #if TRIAL_VERSION
41             Console.WriteLine("Trial mode - 30 days remaining.");
42         #else
43             Console.WriteLine("Full version - thank you for purchasing!");
44         #endif
45
46

```

```

Microsoft Visual Studio Debug Console
Rithika, age 24, grade A
class variable value: 1
App starting...
Trial mode - 30 days remaining.
Maths (4 credits)

C:\Users\ravi.rithika\HelloCohort\HelloCohort\bin\Debug\net10.0\HelloCohort.exe (process 19324) exited with code 0 (0x0)
.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .

```

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```
Course.cs  ModuleA.cs  ModuleB.cs  Program.cs
HelloCohort
SchoolManagement.Course
1 namespace SchoolManagement
2 {
3     2 references
4     public class Course
5     {
6         #region Fields
7         private string _title;
8         private int _credits;
9         #endregion
10    > Properties
14
15    > Constructors
22
23    #region Methods
24    1 reference
25    public void Describe() =>
26        Console.WriteLine($"{_title} ({_credits} credits)");
27    #endregion
28    }
```